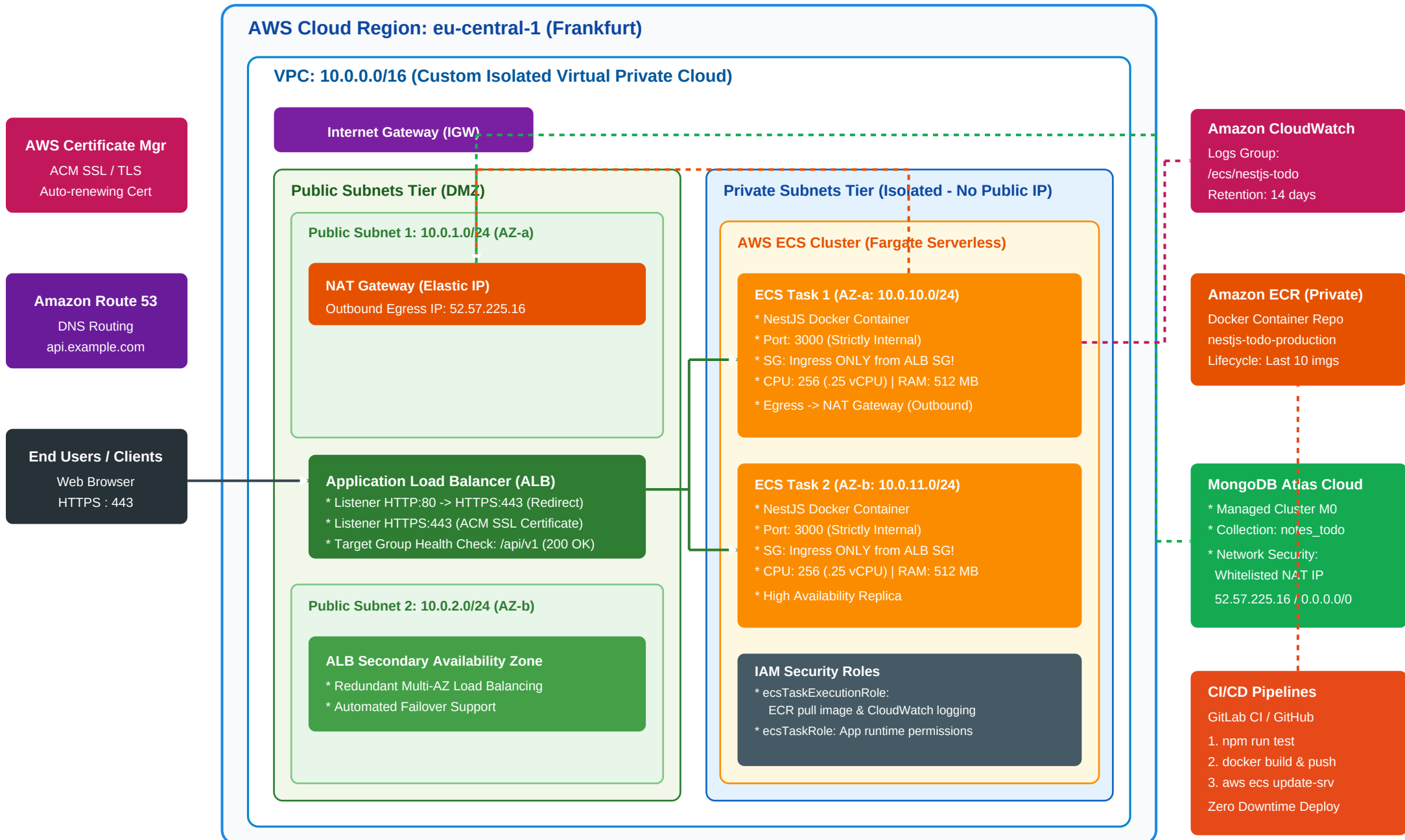


# AWS Production Architecture: NestJS Todo App

High-Availability Serverless Architecture with ECS Fargate, Multi-AZ VPC, ALB & CI/CD

AWS Production Infrastructure

DevOps Architecture Blueprint | Terraform + ECS Fargate



# AWS Infrastructure Specification & Technical Matrix

Detailed Inventory of Provisioned Cloud Resources, Security Policies & CI/CD Stages

| Resource / Component      | AWS Service         | Network Zone / Placement      | Security & Ports                  | Role / Specification                                 |
|---------------------------|---------------------|-------------------------------|-----------------------------------|------------------------------------------------------|
| Virtual Private Cloud     | AWS VPC             | eu-central-1 (10.0.0.0/16)    | DNS Hostnames: Enabled            | Isolated virtual network environment                 |
| Public Subnet 1 & 2       | VPC Subnet          | Multi-AZ: eu-central-1a & 1b  | CIDR: 10.0.1.0/24, 10.0.2.0/24    | Hosts Internet-facing ALB and NAT Gateway            |
| Private Subnet 1 & 2      | VPC Subnet          | Multi-AZ: eu-central-1a & 1b  | CIDR: 10.0.10.0/24, 10.0.11.0/24  | Hosts ECS Tasks (Strictly no Public IP)              |
| Internet Gateway (IGW)    | VPC IGW             | VPC Edge (Attached)           | Ingress / Egress 0.0.0.0/0        | Connects VPC Public subnets to Internet              |
| NAT Gateway               | AWS NAT GW          | Public Subnet 1 (AZ-a)        | Elastic IP: 52.57.225.16          | Provides outbound Internet access for ECS tasks      |
| Application Load Balancer | AWS ALB             | Public Subnets (Multi-AZ)     | SG Ingress: TCP 80 & 443          | Routes traffic, terminates SSL, Health Checks        |
| ALB Target Group          | AWS Target Group    | VPC Internal (IP Target Type) | Health Path: /api/v1 (200 OK)     | Deregistration delay: 30s, health check: 30s         |
| ECS Cluster               | AWS ECS Fargate     | VPC Private Subnets           | Fargate Serverless                | Auto-healing, container management, no EC2 instances |
| ECS Tasks (2 Replicas)    | ECS Task Definition | Private Subnets (Multi-AZ)    | Port 3000 Ingress ONLY from ALB   | NestJS App, CPU: 256, RAM: 512MB, env: DATABASE_URI  |
| Container Registry        | Amazon ECR          | AWS Managed (Private)         | Encrypted, Scan on push           | Stores Docker images with short SHA and latest tags  |
| Log Monitoring            | CloudWatch Logs     | AWS Managed                   | /ecs/nestjs-todo-production       | Centralized stdout/stderr logs, 14 days retention    |
| Security IAM Roles        | AWS IAM             | Global / Attached to ECS      | AmazonECSTaskExecutionRolePolicy  | Grants permissions to pull ECR & write logs          |
| NoSQL Database            | MongoDB Atlas       | Managed External Cloud        | Whitelisted NAT EIP: 52.57.225.16 | M0 Free Sandbox Cluster, DB: notes_todo              |
| CI/CD Pipeline            | GitHub / GitLab     | Cloud Runners (Ubuntu)        | IAM Access Key / Secrets          | 1. npm test -> 2. Docker build/push -> 3. ECS deploy |

Security & Reliability Highlights:

1. Zero Public Exposure: ECS Fargate containers have NO public IP addresses and accept traffic ONLY from the Application Load Balancer Security Group.
2. High Availability: Redundant subnets and containers deployed across 2 independent Availability Zones (eu-central-1a & eu-central-1b) with rolling zero-downtime updates.